

§7.2 - Trigonometric Integrals

Definition #1

The functions tangent and secant are given by

$$\tan(x) = \frac{\sin(x)}{\cos(x)} \quad \text{and} \quad \sec(x) = \frac{1}{\cos(x)}$$

for $x \neq k\pi + \pi/2$ where $k \in \mathbf{Z}$.

Theorem #1

We have the Pythagorean identities

$$\cos^2(x) + \sin^2(x) = 1 \quad \text{and} \quad \tan^2(x) = \sec^2(x) - 1,$$

the power reduction/half-angle identities

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \quad \text{and} \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

and

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x).$$

Theorem #2

We have

$$\frac{d}{dx} \sin(x) = \cos(x) \quad \text{and} \quad \frac{d}{dx} \cos(x) = -\sin(x).$$

Also

$$\int \sin(x) \, dx = -\cos(x) + C \quad \text{and} \quad \int \cos(x) \, dx = \sin(x) + C$$

where $C \in \mathbf{R}$.

Theorem #3

We have

$$\frac{d}{dx} \tan(x) = \sec^2(x) \quad \text{and} \quad \frac{d}{dx} \sec(x) = \sec(x) \cdot \tan(x).$$

Also

$$\int \tan(x) \, dx = \ln |\sec(x)| + C$$

and

$$\int \sec(x) \, dx = \ln |\sec(x) + \tan(x)| + C$$

where $C \in \mathbf{R}$.

Example #1

Find

$$\int \cos^2(x) \, dx.$$

Example #2

Find

$$\int \cos^5(x) \, dx.$$

Example #3

Find

$$\int \sin^3(x) \cos^5(x) \, dx.$$

Example #4

Find

$$\int \sin^2(x) \cos^2(x) \, dx.$$

Idea

$$\int \sin^m(x) \cos^n(x) \, dx$$

1. m odd, use

$$\sin^m(x) = \sin^{m-1}(x) \sin(x) \quad \text{and} \quad \sin^2(x) = 1 - \cos^2(x)$$

2. n odd, use

$$\cos^n(x) = \cos^{n-1}(x) \cos(x) \quad \text{and} \quad \cos^2(x) = 1 - \sin^2(x)$$

3. m and n even, use

$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \quad \text{and} \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

Example #5

Find

$$\int \tan^3(x) \, dx.$$

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integrate tan^3(x)



 Natural Language

 Math Input



Indefinite integral:

☒ Step-by-step solution

$$\int \tan^3(x) dx = \frac{\sec^2(x)}{2} + \log(\cos(x)) + \text{constant}$$

(assuming a complex-valued logarithm)

Example #6

Find

$$\int \sec^4(x) \, dx.$$

Idea

$$\int \tan^m(x) \sec^n(x) \, dx$$

1. m odd, use

$$\tan^2(x) = \sec^2(x) - 1 \quad \text{and} \quad \text{“save” a } \sec(x) \tan(x)$$

2. n even, use

$$\sec^2(x) = 1 + \tan^2(x) \quad \text{and} \quad \text{“save” a } \sec^2(x)$$

3. otherwise, experiment...

Example #7

Find

$$\int \cot(x) \, dx.$$